

Consumer Pulse 360: Integrated Purchase & Sentiment Intelligence Dashboard

An E-Commerce Data Analytics Project

Created by: **Veena V R**

Tools: Power BI, Excel, Python



Project Overview

The Challenge

In today's competitive E-Commerce landscape, understanding what customers buy is only half the story. We need to know *how they feel* about their purchases to drive meaningful growth.

This project bridges the gap between transactional data and customer perception, creating a unified intelligence platform.

Approach Used

Dataset: Comprehensive analysis of 609 customers and 100 products spanning 2023–2024

Goal: Develop an integrated Power BI dashboard that connects sales trends, review sentiment, and customer engagement metrics

Outcome: Actionable insights that align product strategy with customer satisfaction

Revenue Performance at a Glance

₹489K

Total Revenue

Generated from
customer purchases

609

Active
Customers

Engaged purchasers
analyzed

100

Product SKUs

Items in our catalog

₹4.3K

Top Customer
Value

Ashley Parrish's
contribution

Strong revenue performance driven by focused customer engagement and strategic product portfolio management across two high-performing categories.

Category Performance Deep Dive

Home Appliances

53.7% of total revenue

Leading performers: Washing machines and refrigerators driving consistent sales volume

Electronics

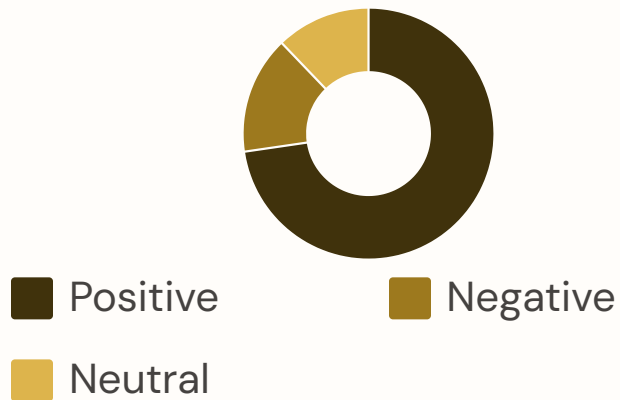
46.3% of total revenue

Star products: Televisions and headphones capturing strong market demand

Our two core categories demonstrate balanced performance, with home appliances showing slight dominance. This distribution reveals opportunities to optimize inventory and marketing spend across both segments.



Customer Sentiment Analysis



What Customers Are Saying

73% positive sentiment reflects strong overall satisfaction, indicating the product selection and customer experience resonate well with buyers.

Average sentiment score: 0.31 – a solid positive indicator that customers are generally pleased with their purchases.

The 15% negative feedback presents a targeted opportunity for improvement, particularly in addressing specific pain points in product quality or delivery experience.

Top Performing Products

Television

Highest unit sales across all categories, driven by competitive pricing and strong customer reviews

Headphones

Consistent performer in electronics segment with excellent sentiment scores

Washing Machine

Top home appliance seller with strong repeat customer recommendations

These three products represent the core revenue drivers and demonstrate the power of combining quality offerings with positive customer perception.



Sales Trends: Seasonal Patterns Revealed

November 2023

Peak sales month driven by holiday shopping season and promotional campaigns

May 2024

Second major peak indicating spring buying patterns and successful marketing initiatives

1

2

3

December 2023 – April 2024

Moderate activity with steady baseline performance

Understanding these cyclical patterns enables proactive inventory management and strategic campaign timing to maximize revenue during high-demand periods.

Business Impact & Strategic Value



Unified Intelligence Platform

Integrated view combining sales performance metrics with customer sentiment analysis, eliminating data silos and enabling holistic decision-making



Strategic Alignment

Direct correlation between customer satisfaction scores and revenue performance guides product development and marketing strategy refinement



Customer-Centric Growth

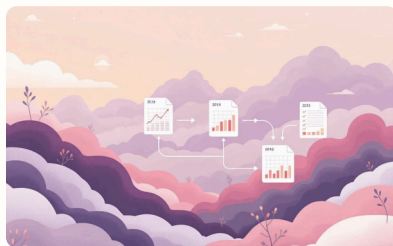
Linking satisfaction directly to financial outcomes creates actionable feedback loops that drive continuous improvement

Technical Implementation



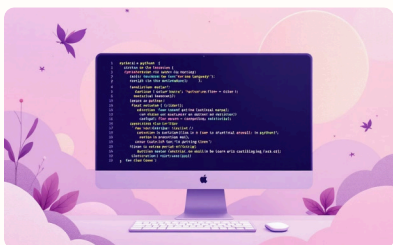
Power BI Analytics

Advanced DAX calculations enable precise KPI tracking, dynamic filtering, and sophisticated trend visualizations, linking sentiment directly to revenue.



Data Engineering

Ensures data quality and consistency through Excel-based cleaning and preparation. Standardized models support scalable and efficient analysis.



Sentiment Analysis

Python-powered NLP extracts and quantifies customer sentiment from review text. Automated scoring provides actionable metrics.



Key Takeaways

The Consumer Pulse 360 Dashboard transforms how we understand our business by connecting the dots between what customers buy and how they feel about their experience.

This integrated analytics solution proves that combining **quantitative sales data** with **qualitative sentiment analysis** unlocks powerful insights that drive customer-focused growth and smarter eCommerce strategy.

By monitoring both performance metrics and customer perception in real-time, organizations can respond faster, optimize better, and grow sustainably.

Veena V R | Data Analyst | Power BI | SQL | Python | Adv Excel

Project Timeline: 2024–2025

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